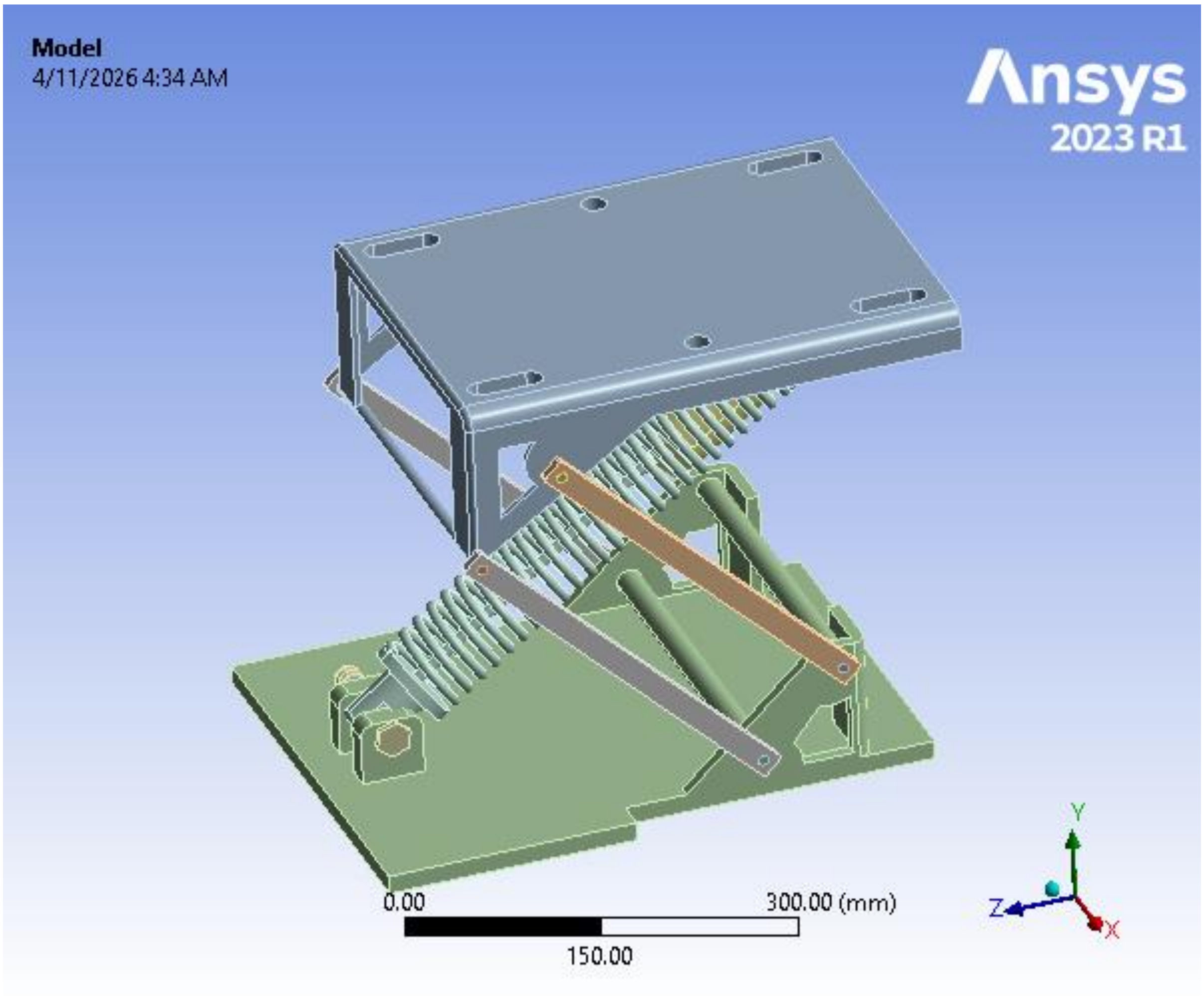
Project*

First Saved	Saturday, April 11, 2026
Last Saved	Saturday, April 11, 2026
Product Version	2023 R1
Save Project Before Solution	No
Save Project After Solution	No



Contents

Report Not Finalized

Not all objects described below are in a finalized state. As a result, data may be incomplete, obsolete or in error. View first state problem. To finalize this report, edit objects as needed and solve the analyses.

Units

TABLE 1

Unit System	Metric (mm, kg, N, s, mV, mA) Degrees rad/s Celsius
Angle	Degrees
Rotational Velocity	rad/s
Temperature	Celsius

Model (A4)

TABLE 2
Model (A4) > Geometry Imports

Object Name	*Geometry Imports*
State	Solved

TABLE 3
Model (A4) > Geometry Imports > Geometry Import (A3)

Object Name	*Geometry Import (A3)*
State	Solved
Definition	
Source	C:\Users\cclab1\Pictures\dme project\Assembly Seat stp.STEP
Type	Step
Basic Geometry Options	
Solid Bodies	Yes
Surface Bodies	Yes
Line Bodies	No
Parameters	Independent
Parameter Key	ANS;DS
Attributes	No
Named Selections	No
Material Properties	No
Advanced Geometry Options	
Use Associativity	Yes
Coordinate Systems	No
Reader Mode Saves Updated File	No
Use Instances	Yes
Smart CAD Update	Yes
Compare Parts On Update	No
Analysis Type	3-D
Mixed Import Resolution	None
Import Facet Quality	Source
Clean Bodies On Import	No
Stitch Surfaces On Import	None
Decompose Disjoint Geometry	Yes
Enclosure and Symmetry Processing	Yes

Geometry

TABLE 4
Model (A4) > Geometry

Object Name	*Geometry*
State	Fully Defined
Definition	
Source	C:\Users\cclab1\Pictures\dme project\Assembly Seat stp.STEP
Type	Step
Length Unit	Millimeters
Element Control	Program Controlled
Display Style	Body Color
Bounding Box	
Length X	350. mm
Length Y	385.38 mm
Length Z	487.21 mm
Properties	
Volume	5.702e+006 mm^3
Mass	44.843 kg
Scale Factor Value	1.

Statistics	
Bodies	21
Active Bodies	21
Nodes	156754
Elements	43802
Mesh Metric	None
Update Options	
Assign Default Material	No
Basic Geometry Options	
Solid Bodies	Yes
Surface Bodies	Yes
Line Bodies	No
Parameters	Independent
Parameter Key	ANS;DS
Attributes	No
Named Selections	No
Material Properties	No
Advanced Geometry Options	
Use Associativity	Yes
Coordinate Systems	No
Reader Mode Saves Updated File	No
Use Instances	Yes
Smart CAD Update	Yes
Compare Parts On Update	No
Analysis Type	3-D
Mixed Import Resolution	None
Import Facet Quality	Source
Clean Bodies On Import	No
Stitch Surfaces On Import	None
Decompose Disjoint Geometry	Yes
Enclosure and Symmetry Processing	Yes

TABLE 5
Model (A4) > Geometry > Parts

scissor part|Fillet2	*top part 1|Extrude-Thin8*	*top part modifies|Boss-Extrude6*	*scissor part|Fillet2[2]*	*bootom spring|Boss-Extrude5*	*head part of spring|Thread1*	*part more|Cut-Extrude3*	*bolt|Thread1*	*bolt|Thread1[2]*	*nut|Cu Extrud*
Meshed									
Graphics Properties									
Yes									
1									
Definition									
No									
Flexible									
Default Coordinate System									
By Environment									
None									
Material									

Low alloy steel, 4340, normalized				316 Stainless Steel			Low alloy steel, 4340, normalized		Stru
Yes									
Yes									
Bounding Box									
8. mm	300. mm	350. mm	8. mm	81.234 mm	80. mm	103.03 mm	105. mm		
247.49 mm	136.73 mm	115. mm	247.49 mm	320.36 mm	179.73 mm	89.777 mm	31.475 mm		2
247.49 mm	400.07 mm	450. mm	247.49 mm	390.15 mm	202.44 mm	72.263 mm	32.242 mm		3
Properties									
51950 mm³	1.9905e+006 mm³	2.6527e+006 mm³	51950 mm³	3.6377e+005 mm³	4.0066e+005 mm³	24864 mm³	27233 mm³		19
0.40781 kg	15.625 kg	20.824 kg	0.40781 kg	2.8935 kg	3.1869 kg	0.19777 kg	0.21378 kg		1.53
931.14 mm	777.36 mm	777.34 mm	931.14 mm	777.29 mm	777.34 mm	777.47 mm	771.31 mm		7
969.61 mm	1196.4 mm	842.45 mm	1034.6 mm	985.38 mm	1117.6 mm	1127.9 mm	876.2 mm	1170.1 mm	876.2 mm
1646.2 mm	1567.3 mm	1615.6 mm	1578.7 mm	1646.3 mm	1470. mm	1455.8 mm	1792.3 mm	1399.9 mm	1792. mm
3604.6 kg·mm²	2.2794e+005 kg·mm²	3.7822e+005 kg·mm²	3604.6 kg·mm²	49880 kg·mm²	9636.9 kg·mm²	86.336 kg·mm²	220.13 kg·mm²		0.88
15.944 kg·mm²	3.5173e+005 kg·mm²	5.6053e+005 kg·mm²	15.944 kg·mm²	2356.8 kg·mm²	9587.8 kg·mm²	129.47 kg·mm²	220.13 kg·mm²		0.88
3616.2 kg·mm²	1.345e+005 kg·mm²	2.037e+005 kg·mm²	3616.2 kg·mm²	49868 kg·mm²	1309.8 kg·mm²	212.51 kg·mm²	10.021 kg·mm²		1.7
Statistics									
525	8732	5268	525	9146	6226	1192	3823		
59	4229	2507	59	3665	3360	609	1990		
None									

TABLE 6
Model (A4) > Geometry > Parts

Object Name	*drive shaft pin|drive shaft pin chamfer*	*drive shaft pin|drive shaft pin chamfer[2]*	*drive shaft pin|drive shaft pin chamfer[3]*	*drive shaft pin|drive shaft pin chamfer[4]*	*scissor part|Fillet2[3]*	*scissor part|Fillet2[4]*	*drive shaft pin|drive shaft pin chamfer[5]*	*drive shaft pin|drive shaft pin chamfer[6]*	*drive shaft pin|drive shaft pin chamfer[7]*	*drive shaft pin|drive shaft pin chamfer[8]*
State	Meshed									
Graphics Properties										
Visible	Yes									
Transparency	1									
Definition										
Suppressed	No									
Stiffness Behavior	Flexible									
Coordinate System	Default Coordinate System									
Reference Temperature	By Environment									

Treatment	None									
Material										
Assignment	Structural Steel				Low alloy steel, 4340, normalized		Structural Steel			
Nonlinear Effects	Yes									
Thermal Strain Effects	Yes									
Bounding Box										
Length X	8. mm									
Length Y	8.8788 mm				247.49 mm		8.8788 mm	10.995 mm	8.8788 mm	11.49 mm
Length Z	8.8788 mm				247.49 mm		8.8788 mm	10.995 mm	8.8788 mm	11.49 mm
Properties										
Volume	413.16 mm^3				51950 mm^3		413.16 mm^3			
Mass	3.2433e-003 kg				0.40781 kg		3.2433e-003 kg			
Centroid X	931.24 mm				623.54 mm		623.44 mm			
Centroid Y	1078.5 mm	860.71 mm	1143.5 mm	925.71 mm	1034.6 mm	969.61 mm	925.71 mm	860.71 mm	1143.5 mm	1078.5 mm
Centroid Z	1755.1 mm	1537.3 mm	1687.6 mm	1469.8 mm	1578.7 mm	1646.2 mm	1469.8 mm	1537.3 mm	1687.6 mm	1755.1 mm
Moment of Inertia Ip1	3.0606e-002 kg·mm^2				3604.6 kg·mm^2		3.0606e-002 kg·mm^2			
Moment of Inertia Ip2	3.0606e-002 kg·mm^2				15.944 kg·mm^2		3.0606e-002 kg·mm^2			
Moment of Inertia Ip3	2.6659e-002 kg·mm^2				3616.2 kg·mm^2		2.6659e-002 kg·mm^2			
Statistics										
Nodes	14471				525		14471			
Elements	3141				59		3141			
Mesh Metric	None									

TABLE 7
Model (A4) > Materials

Object Name	*Materials*
State	Fully Defined
Statistics	
Materials	3
Material Assignments	2

TABLE 8
Model (A4) > Materials > Low alloy steel, 4340, normalized Assignment

Object Name	*Low alloy steel, 4340, normalized Assignment*	*316 Stainless Steel Assignment*
State	Fully Defined	
General		
Scoping Method	Geometry Selection	
Geometry	8 Bodies	3 Bodies
Definition		
Material Name	Low alloy steel, 4340, normalized	316 Stainless Steel
Nonlinear Effects	Yes	
Thermal Strain Effects	Yes	

Reference Temperature	By Environment
Suppressed	No

Coordinate Systems

TABLE 9
Model (A4) > Coordinate Systems > Coordinate System

Object Name	*Global Coordinate System*
State	Fully Defined
Definition	
Type	Cartesian
Coordinate System ID	0.
Origin	
Origin X	0. mm
Origin Y	0. mm
Origin Z	0. mm
Directional Vectors	
X Axis Data	[1. 0. 0.]
Y Axis Data	[0. 1. 0.]
Z Axis Data	[0. 0. 1.]

Connections

TABLE 10
Model (A4) > Connections

Object Name	*Connections*
State	Fully Defined
Auto Detection	
Generate Automatic Connection On Refresh	Yes
Transparency	
Enabled	Yes
Statistics	
Contacts	32
Active Contacts	32
Joints	0
Active Joints	0
Beams	0
Active Beams	0
Bearings	0
Active Bearings	0
Springs	0
Active Springs	0
Body Interactions	0
Active Body Interactions	0

TABLE 11
Model (A4) > Connections > Contacts

Object Name	*Contacts*
State	Fully Defined
Definition	
Connection Type	Contact

Scope	
Scoping Method	Geometry Selection
Geometry	All Bodies
Auto Detection	
Tolerance Type	Slider
Tolerance Slider	0.
Tolerance Value	1.7825 mm
Use Range	No
Face/Face	Yes
Face-Face Angle Tolerance	75. °
Face Overlap Tolerance	Off
Cylindrical Faces	Include
Face/Edge	No
Edge/Edge	No
Priority	Include All
Group By	Bodies
Search Across	Bodies
Statistics	
Connections	32
Active Connections	32

TABLE 12
Model (A4) > Connections > Contacts > Contact Regions

ct e	*Contact Region*	*Contact Region 2*	*Contact Region 3*	*Contact Region 4*	*Contact Region 5*	*Contact Region 6*	*Contact Region 7*	*Contact Region 8*	*Contact Region 9*	*Contact Region 10*	*F*
te	Fully Defined										
Scope											
g d	Geometry Selection										
ct	3 Faces	1 Face	2 Faces		1 Face	4 Faces	5 Faces	2 Faces			
et	4 Faces	1 Face	2 Faces		1 Face	3 Faces	10 Faces	1 Face	2 Faces		
ct s	scissor part|Fillet2				top part 1|Extrude-Thin8						
et s	top part 1|Extrude-Thin8	top part modifies|Boss-Extrude6	drive shaft pin|drive shaft pin chamfer	drive shaft pin|drive shaft pin chamfer[2]	scissor part|Fillet2[2]	head part of spring|Thread1	bolt|Thread1[2]	nut|Cut-Extrude1[2]	drive shaft pin|drive shaft pin chamfer	drive shaft pin|drive shaft pin chamfer[3]	pa
d	No										
Definition											
e	Bonded										
e e	Automatic										
or	Program Controlled										
m ct	Program Controlled										
m e	1.7825 mm										
ed	No										
Display											
nt s	No										

	Advanced
n	Program Controlled
ll g	Program Controlled
n d	Program Controlled
n e	Program Controlled
p e	Program Controlled
al s	Program Controlled
e s	Program Controlled
ll n	Program Controlled
	Geometric Modification
ct y n	None
et y n	None

TABLE 13
Model (A4) > Connections > Contacts > Contact Regions

ect me	*Contact Region 12*	*Contact Region 13*	*Contact Region 14*	*Contact Region 15*	*Contact Region 16*	*Contact Region 17*	*Contact Region 18*	*Contact Region 19*	*Contact Region 20*	*Contact Region 21*	*F*
ate	Fully Defined										
	Scope										
ng od	Geometry Selection										
act	3 Faces	2 Faces		5 Faces	1 Face	2 Faces			1 Face	2 Fac	
get	2 Faces	1 Face	2 Faces	10 Faces	1 Face	2 Faces		1 Face		2 Fac	
act es	top part 1|Extrude-Thin8	top part modifies|Boss-Extrude6								scissor part	
get es	scissor part|Fillet2[4]	scissor part|Fillet2[2]	bootom spring|Boss-Extrude5	bolt|Thread1	nut|Cut-Extrude1	drive shaft pin|drive shaft pin chamfer[2]	drive shaft pin|drive shaft pin chamfer[4]	scissor part|Fillet2[3]	scissor part|Fillet2[4]	drive shaft pin|drive shaft pin chamfer[3]	d c
ed	No										
	Definition										
pe	Bonded										
pe de	Automatic										
ior	Program Controlled										
im act	Program Controlled										
im ce	1.7825 mm										
ed	No										
	Display										

ent als	No
Advanced	
on	Program Controlled
nall ng	Program Controlled
on od	Program Controlled
on ce	Program Controlled
lip ce	Program Controlled
nal ess	Program Controlled
ate ess	Program Controlled
all on	Program Controlled
Geometric Modification	
act try on	None
get try on	None

TABLE 14
Model (A4) > Connections > Contacts > Contact Regions

Object Name	*Contact Region 23*	*Contact Region 24*	*Contact Region 25*	*Contact Region 26*	*Contact Region 27*	*Contact Region 28*	*Contact Region 29*	*Contact Region 30*	*Contact Region 31*	*Co Regi*
State	Fully Defined									
Scope										
oping ethod	Geometry Selection									
ontact	2 Faces				1 Face		2 Faces			
Target	8 Faces	6 Faces	2 Faces	7 Faces	2 Faces					
ontact odies	bootom spring|Boss-Extrude5		head part of spring|Thread1		bolt|Thread1	bolt|Thread1[2]	scissor part|Fillet2[3]		scissor part|Fill	
Target odies	head part of spring|Thread1	bolt|Thread1	part more|Cut-Extrude3	bolt|Thread1[2]	nut|Cut-Extrude1	nut|Cut-Extrude1[2]	drive shaft pin|drive shaft pin chamfer[5]	drive shaft pin|drive shaft pin chamfer[7]	drive shaft pin|drive shaft pin chamfer[6]	drive pin| sha cham
tected	No									
Definition										
Type	Bonded									
Scope Mode	Automatic									
havior	Program Controlled									
Trim ontact	Program Controlled									
Trim erance	1.7825 mm									
essed	No									
Display										

ement ormals	No
Advanced	
ulation	Program Controlled
Small Sliding	Program Controlled
ection ethod	Program Controlled
tration erance	Program Controlled
c Slip erance	Program Controlled
lormal fness	Program Controlled
pdate fness	Program Controlled
Pinball Region	Program Controlled
Geometric Modification	
ontact metry ection	None
Target metry ection	None

Mesh

TABLE 15
Model (A4) > Mesh

Object Name	*Mesh*
State	Solved
Display	
Display Style	Use Geometry Setting
Defaults	
Physics Preference	Mechanical
Element Order	Program Controlled
Element Size	Default
Sizing	
Use Adaptive Sizing	Yes
Resolution	Default (2)
Mesh Defeaturing	Yes
Defeature Size	Default
Transition	Fast
Span Angle Center	Medium
Initial Size Seed	Assembly
Bounding Box Diagonal	713.02 mm
Average Surface Area	2409.8 mm²
Minimum Edge Length	0.14142 mm
Quality	
Check Mesh Quality	Yes, Errors
Error Limits	Aggressive Mechanical
Target Element Quality	Default (5.e-002)
Smoothing	Medium

Mesh Metric	None
Inflation	
Use Automatic Inflation	None
Inflation Option	Smooth Transition
Transition Ratio	0.272
Maximum Layers	5
Growth Rate	1.2
Inflation Algorithm	Pre
View Advanced Options	No
Advanced	
Number of CPUs for Parallel Part Meshing	Program Controlled
Straight Sided Elements	No
Rigid Body Behavior	Dimensionally Reduced
Triangle Surface Mesher	Program Controlled
Topology Checking	Yes
Pinch Tolerance	Please Define
Generate Pinch on Refresh	No
Statistics	
Nodes	156754
Elements	43802
Show Detailed Statistics	No

TABLE 16
Model (A4) > Mesh > Mesh Controls

Object Name	*Body Sizing*
State	Fully Defined
Scope	
Scoping Method	Geometry Selection
Geometry	11 Bodies
Definition	
Suppressed	No
Type	Element Size
Element Size	Default (35.539 mm)
Advanced	
Defeature Size	Default
Behavior	Soft

Static Structural (A5)

TABLE 17
Model (A4) > Analysis

Object Name	*Static Structural (A5)*
State	Not Solved
Definition	
Physics Type	Structural
Analysis Type	Static Structural
Solver Target	Mechanical APDL
Options	
Environment Temperature	22. °C
Generate Input Only	No

TABLE 18
Model (A4) > Static Structural (A5) > Analysis Settings

Object Name	*Analysis Settings*
State	Fully Defined
Restart Analysis	
Restart Type	Program Controlled
Load Step	1
Substep	4
Time	7.1875e-002 s
Step Controls	
Number Of Steps	1.
Current Step Number	1.
Step End Time	1. s
Auto Time Stepping	Program Controlled
Solver Controls	
Solver Type	Program Controlled
Weak Springs	Off
Solver Pivot Checking	Program Controlled
Large Deflection	Off
Inertia Relief	Off
Quasi-Static Solution	Off
Rotordynamics Controls	
Coriolis Effect	Off
Restart Controls	
Generate Restart Points	Program Controlled
Retain Files After Full Solve	No
Combine Restart Files	Program Controlled
Nonlinear Controls	
Newton-Raphson Option	Program Controlled
Force Convergence	Program Controlled
Moment Convergence	Program Controlled
Displacement Convergence	Program Controlled
Rotation Convergence	Program Controlled
Line Search	Program Controlled
Stabilization	Program Controlled
Advanced	
Inverse Option	No

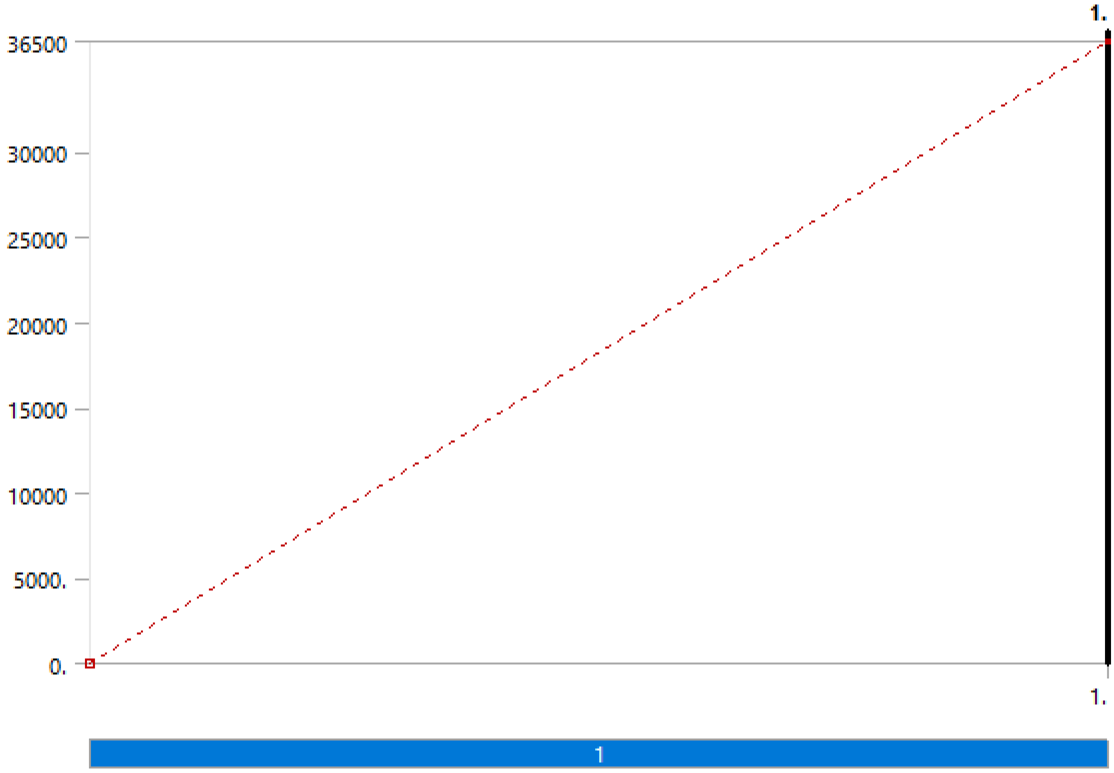
Contact Split (DMP)	Off
Output Controls	
Stress	Yes
Back Stress	No
Strain	Yes
Contact Data	Yes
Nonlinear Data	No
Nodal Forces	No
Volume and Energy	Yes
Euler Angles	Yes
General Miscellaneous	No
Contact Miscellaneous	No
Store Results At	All Time Points
Result File Compression	Program Controlled
Analysis Data Management	
Solver Files Directory	C:\Users\cclab1\AppData\Local\Temp\WB_cclab1_59300_2\wbnew_files\dp0\SYS\MECH\
Future Analysis	None
Scratch Solver Files Directory	
Save MAPDL db	No
Contact Summary	Program Controlled
Delete Unneeded Files	Yes
Nonlinear Solution	Yes
Solver Units	Active System
Solver Unit System	nmm

TABLE 19
Model (A4) > Static Structural (A5) > Loads

Object Name	*Force*	*Fixed Support*
State	Fully Defined	
Scope		
Scoping Method	Geometry Selection	
Geometry	1 Face	
Definition		
Type	Force	Fixed Support
Define By	Vector	
Applied By	Surface Effect	
Magnitude	36500 N (ramped)	
Direction	Defined	

Suppressed	No

FIGURE 1
Model (A4) > Static Structural (A5) > Force



Solution (A6)

TABLE 20
Model (A4) > Static Structural (A5) > Solution

Object Name	*Solution (A6)*
State	Solve Failed
Adaptive Mesh Refinement	
Max Refinement Loops	1.
Refinement Depth	2.
Information	
Status	Solve Required, Restart Available
MAPDL Elapsed Time	10 m 38 s
MAPDL Memory Used	4.9658 GB
MAPDL Result File Size	145.31 MB
Post Processing	
Beam Section Results	No
On Demand Stress/Strain	No

TABLE 21
Model (A4) > Static Structural (A5) > Solution (A6) > Solution Information

Object Name	*Solution Information*
State	Solve Failed
Solution Information	
Solution Output	Solver Output
Newton-Raphson Residuals	0

Identify Element Violations	0
Update Interval	2.5 s
Display Points	All
FE Connection Visibility	
Activate Visibility	Yes
Display	All FE Connectors
Draw Connections Attached To	All Nodes
Line Color	Connection Type
Visible on Results	No
Line Thickness	Single
Display Type	Lines

TABLE 22
Model (A4) > Static Structural (A5) > Solution (A6) > Results

Object Name	*Total Deformation*	*Equivalent Elastic Strain*	*Equivalent Stress*	*Strain Energy*
State	Solve Failed			
Scope				
Scoping Method	Geometry Selection			
Geometry	All Bodies			
Definition				
Type	Total Deformation	Equivalent Elastic Strain	Equivalent (von-Mises) Stress	Strain Energy
By	Time			
Display Time	Last			
Separate Data by Entity	No			
Calculate Time History	Yes			
Identifier				
Suppressed	No			
Results				
Minimum				
Maximum				
Average				
Minimum Occurs On				
Maximum Occurs On				
Total				
Information				
Time				
Load Step	0			
Substep	0			
Iteration Number				
Integration Point Results				
Display Option		Averaged		
Average Across Bodies		No		

TABLE 23
Model (A4) > Static Structural (A5) > Solution (A6) > Fatigue Tools

Object Name	*Fatigue Tool*
State	Solve Failed

Domain	
Domain Type	Time
Materials	
Fatigue Strength Factor (Kf)	1.
Loading	
Type	Fully Reversed
Scale Factor	1.
Definition	
Display Time	End Time
Options	
Analysis Type	Stress Life
Mean Stress Theory	None
Stress Component	Equivalent (von-Mises)
Life Units	
Units Name	cycles
1 cycle is equal to	1. cycles

FIGURE 2
Model (A4) > Static Structural (A5) > Solution (A6) > Fatigue Tool

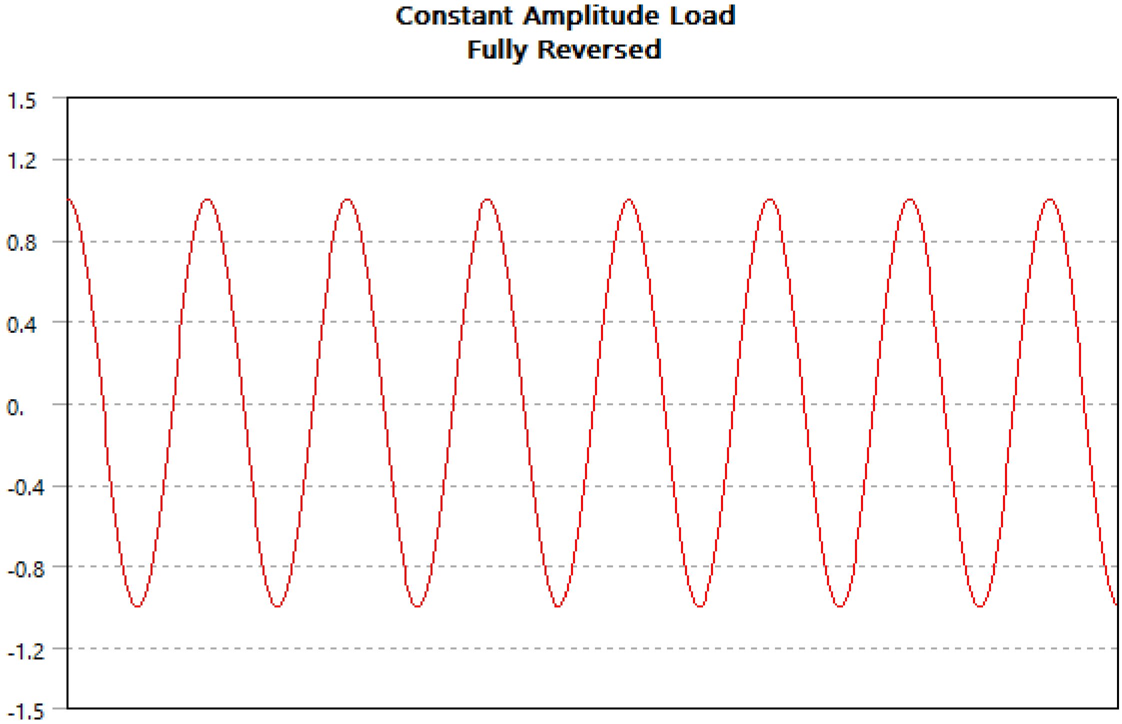


FIGURE 3
Model (A4) > Static Structural (A5) > Solution (A6) > Fatigue Tool

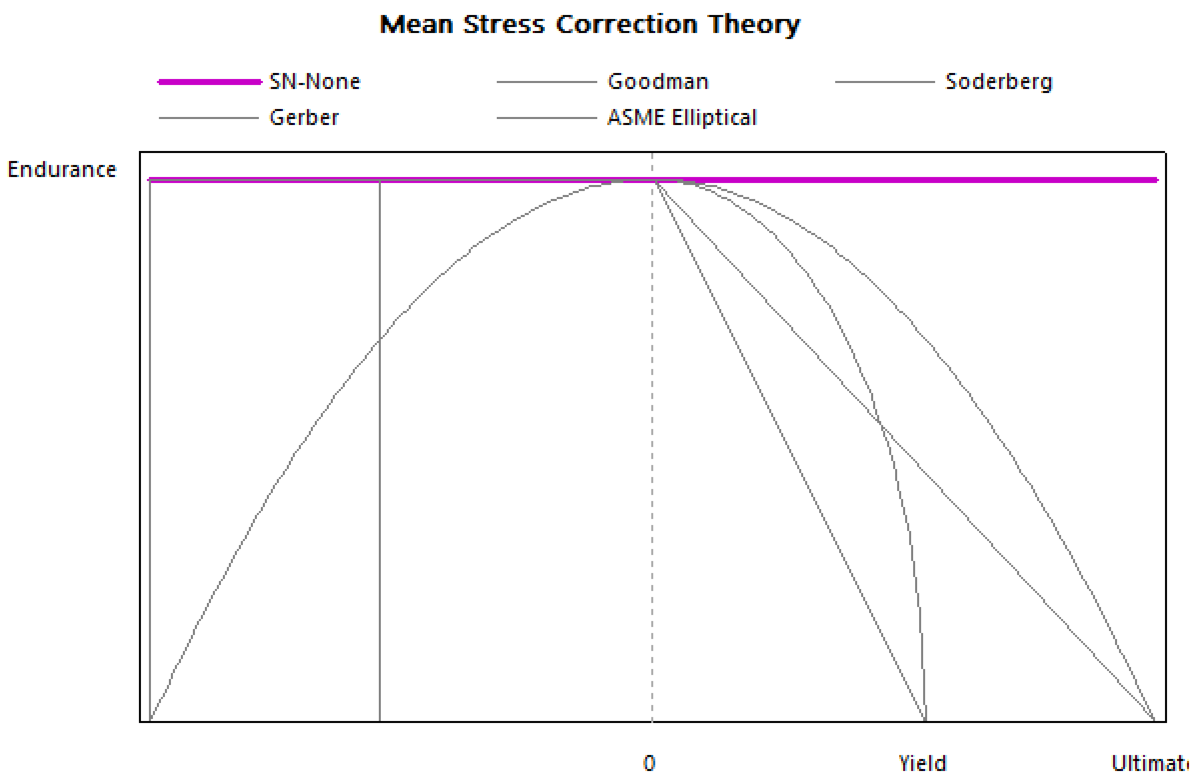


TABLE 24
Model (A4) > Static Structural (A5) > Solution (A6) > Fatigue Tool > Results

Object Name	*Life*	*Damage*	*Safety Factor*	*Equivalent Alternating Stress*
State	Solve Failed			
Scope				
Scoping Method	Geometry Selection			
Geometry	All Bodies			
Definition				
Type	Life	Damage	Safety Factor	Equivalent Alternating Stress
Identifier				
Suppressed	No			
Design Life		1.e+009 cycles		
Results				
Minimum				
Minimum Occurs On				
Maximum				
Maximum Occurs On				
Average				

Material Data

Low alloy steel, 4340, normalized

TABLE 25
Low alloy steel, 4340, normalized > Constants

Density	7.85e-006 kg mm^-3
Tensile Yield Strength	855.3 MPa
Tensile Ultimate Strength	1273 MPa

Specific Heat Constant Pressure	4.783e+005 mJ kg^-1 C^-1
Isotropic Resistivity	2.205e-004 ohm mm

TABLE 26
Low alloy steel, 4340, normalized > Appearance

Red	Green	Blue
204	204	204
Opacity		
1		
Metallic Finish		
1		

TABLE 27
Low alloy steel, 4340, normalized > Isotropic Elasticity

Young's Modulus MPa	Poisson's Ratio	Bulk Modulus MPa	Shear Modulus MPa	Temperature C
2.218e+005	0.29	1.7603e+005	85969	-260.2
2.194e+005	0.29	1.7413e+005	85039	-154.4
2.151e+005	0.29	1.7071e+005	83372	-48.72
2.112e+005	0.29	1.6762e+005	81860	56.99
2.064e+005	0.29	1.6381e+005	80000	162.7
1.992e+005	0.29	1.581e+005	77209	268.4
1.896e+005	0.29	1.5048e+005	73488	374.1
1.776e+005	0.29	1.4095e+005	68837	479.9
1.633e+005	0.29	1.296e+005	63295	585.6
1.466e+005	0.29	1.1635e+005	56822	691.3
1.299e+005	0.29	1.031e+005	50349	797
1.228e+005	0.29	97460	47597	902.7
1.156e+005	0.29	91746	44806	1008
1.085e+005	0.29	86111	42054	1114
1.013e+005	0.29	80397	39264	1220

TABLE 28
Low alloy steel, 4340, normalized > Bilinear Isotropic Hardening

Yield Strength MPa	Tangent Modulus MPa	Temperature C
855.3	4270	23

TABLE 29
Low alloy steel, 4340, normalized > S-N Curve

Alternating Stress MPa	Cycles	R-Ratio
903	1000	-1
839.1	3162	-1
780.8	10000	-1
727.2	31620	-1
677.5	1.e+005	-1
631.4	3.162e+005	-1
588.5	1.e+006	-1
548.6	3.162e+006	-1
511.3	1.e+007	-1
476.7	3.162e+007	-1
444.3	1.e+008	-1
527	1000	0
504.5	3162	0
482.8	10000	0

461.7	31620	0
441.2	1.e+005	0
421.1	3.162e+005	0
401.6	1.e+006	0
382.5	3.162e+006	0
364	1.e+007	0
346.1	3.162e+007	0
328.7	1.e+008	0
288.1	1000	0.5
281.2	3162	0.5
274.3	10000	0.5
267.4	31620	0.5
260.3	1.e+005	0.5
253.2	3.162e+005	0.5
246	1.e+006	0.5
238.7	3.162e+006	0.5
231.3	1.e+007	0.5
223.9	3.162e+007	0.5
216.5	1.e+008	0.5

TABLE 30
Low alloy steel, 4340, normalized > Isotropic Secant Coefficient of Thermal Expansion

Coefficient of Thermal Expansion C^-1	Temperature C
6.869e-006	-270.1
7.585e-006	-236.6
8.258e-006	-203.1
8.887e-006	-169.6
9.474e-006	-136
1.002e-005	-102.5
1.052e-005	-68.97
1.099e-005	-35.44
1.142e-005	-1.915
1.181e-005	31.61
1.216e-005	65.14
1.247e-005	98.67
1.275e-005	132.2
1.3e-005	165.7
1.321e-005	199.3
1.34e-005	232.8
1.354e-005	266.3
1.366e-005	299.9
1.375e-005	333.4
1.381e-005	366.9
1.384e-005	400.4
1.385e-005	434
1.383e-005	467.5
1.378e-005	501
1.371e-005	534.6
1.361e-005	568.1
1.349e-005	601.6
1.335e-005	635.1
1.319e-005	668.7

1.294e-005	702.2
1.128e-005	735.7
1.079e-005	769.3
1.081e-005	802.8
1.108e-005	836.3
1.158e-005	869.9
Zero-Thermal-Strain Reference Temperature C	
20	

TABLE 31
Low alloy steel, 4340, normalized > Isotropic Thermal Conductivity

Thermal Conductivity W mm^-1 C^-1	Temperature C
2.671e-002	-150.2
3.176e-002	-73.52
3.521e-002	3.156
3.731e-002	79.83
3.828e-002	156.5
3.837e-002	233.2
3.781e-002	309.9
3.685e-002	386.5
3.572e-002	463.2
3.466e-002	539.9

TABLE 32
Low alloy steel, 4340, normalized > Johnson Cook Strength

Initial Yield Stress MPa	Hardening Constant MPa	Hardening Exponent	Strain Rate Constant	Thermal Softening Exponent	Melting Temperature C	Reference Strain Rate (/sec)
7.92e+008	5.1e+008	0.26	1.4e-002	1.03	1520	1

316 Stainless Steel

TABLE 33
316 Stainless Steel > Appearance

Red	Green	Blue
161	209	255

TABLE 34
316 Stainless Steel > Bilinear Isotropic Hardening

Yield Strength MPa	Tangent Modulus MPa	Temperature C
489	1001	20
388	691	200
348	670	400
289	634	600
246	388	700

TABLE 35
316 Stainless Steel > Isotropic Elasticity

Young's Modulus MPa	Poisson's Ratio	Bulk Modulus MPa	Shear Modulus MPa	Temperature C
1.95e+005	0.25	1.3e+005	78000	20
1.91e+005	0.26	1.3264e+005	75794	100
1.86e+005	0.275	1.3778e+005	72941	200
1.8e+005	0.315	1.6216e+005	68441	300

1.73e+005	0.33	1.6961e+005	65038	400
1.64e+005	0.3	1.3667e+005	63077	500
1.55e+005	0.32	1.4352e+005	58712	600
1.44e+005	0.31	1.2632e+005	54962	700
1.31e+005	0.24	83974	52823	800
1.17e+005	0.24	75000	47177	900
1.e+005	0.24	64103	40323	1000
81000	0.24	51923	32661	1100
51000	0.24	32692	20565	1200

TABLE 36
316 Stainless Steel > Density

Density kg mm^-3	Temperature C
7.954e-006	26.85
7.91e-006	126.85
7.864e-006	226.85
7.818e-006	326.85
7.771e-006	426.85
7.723e-006	526.85
7.674e-006	626.85
7.624e-006	726.85
7.574e-006	826.85
7.523e-006	926.85
7.471e-006	1026.8
7.419e-006	1126.8
7.365e-006	1226.8
7.311e-006	1326.8
6.979e-006	1426.8
6.92e-006	1526.8
6.857e-006	1626.8
6.791e-006	1726.8
6.721e-006	1826.8
6.648e-006	1926.8
6.571e-006	2026.8
6.49e-006	2126.8
6.406e-006	2226.8
6.318e-006	2326.8
6.229e-006	2426.8
6.131e-006	2526.8
6.032e-006	2626.8
5.93e-006	2726.8

TABLE 37
316 Stainless Steel > Isotropic Secant Coefficient of Thermal Expansion

Coefficient of Thermal Expansion C^-1	Temperature C
1.457e-005	-0.15
1.478e-005	26.85
1.518e-005	76.85
1.561e-005	126.85
1.633e-005	226.85
1.691e-005	326.85
1.742e-005	426.85

1.789e-005	526.85
1.832e-005	626.85
1.871e-005	726.85
1.903e-005	826.85
1.927e-005	926.85
1.945e-005	1026.8
1.961e-005	1126.8
1.977e-005	1226.8
1.99e-005	1326.8
Zero-Thermal-Strain Reference Temperature C	
19.85	

TABLE 38
316 Stainless Steel > Isotropic Thermal Conductivity

Thermal Conductivity W mm^-1 C^-1	Temperature C
1.297e-002	-0.15
1.331e-002	19.85
1.344e-002	26.85
1.432e-002	76.85
1.516e-002	126.85
1.68e-002	226.85
1.836e-002	326.85
1.987e-002	426.85
2.139e-002	526.85
2.279e-002	626.85
2.406e-002	726.85
2.546e-002	826.85
2.674e-002	926.85
2.802e-002	1026.8
2.932e-002	1126.8
3.061e-002	1226.8
3.186e-002	1326.8
3.241e-002	1370.8
2.69e-002	1398.8
2.724e-002	1426.8

TABLE 39
316 Stainless Steel > Specific Heat Constant Pressure

Specific Heat mJ kg^-1 C^-1	Temperature C
4.9873e+005	26.85
5.1212e+005	126.85
5.2551e+005	226.85
5.3848e+005	326.85
5.5187e+005	426.85
5.6526e+005	526.85
5.7865e+005	626.85
5.9162e+005	726.85
6.0501e+005	826.85
6.184e+005	926.85
6.3178e+005	1026.8
6.4475e+005	1126.8
6.5814e+005	1226.8

6.7153e+005	1326.8
7.6986e+005	1426.8
7.6986e+005	1526.8

TABLE 40
316 Stainless Steel > Melting Temperature

Melting Temperature C
1370

Structural Steel

TABLE 41
Structural Steel > Constants

Density	7.85e-006 kg mm^-3
Coefficient of Thermal Expansion	1.2e-005 C^-1
Specific Heat	4.34e+005 mJ kg^-1 C^-1
Thermal Conductivity	6.05e-002 W mm^-1 C^-1
Resistivity	1.7e-004 ohm mm

TABLE 42
Structural Steel > Color

Red	Green	Blue
132	139	179

TABLE 43
Structural Steel > Compressive Ultimate Strength

Compressive Ultimate Strength MPa
0

TABLE 44
Structural Steel > Compressive Yield Strength

Compressive Yield Strength MPa
250

TABLE 45
Structural Steel > Tensile Yield Strength

Tensile Yield Strength MPa
250

TABLE 46
Structural Steel > Tensile Ultimate Strength

Tensile Ultimate Strength MPa
460

TABLE 47
Structural Steel > Isotropic Secant Coefficient of Thermal Expansion

Zero-Thermal-Strain Reference Temperature C
22

TABLE 48
Structural Steel > S-N Curve

Alternating Stress MPa	Cycles	Mean Stress MPa
3999	10	0

2827	20	0
1896	50	0
1413	100	0
1069	200	0
441	2000	0
262	10000	0
214	20000	0
138	1.e+005	0
114	2.e+005	0
86.2	1.e+006	0

TABLE 49
Structural Steel > Strain-Life Parameters

Strength Coefficient MPa	Strength Exponent	Ductility Coefficient	Ductility Exponent	Cyclic Strength Coefficient MPa	Cyclic Strain Hardening Exponent
920	-0.106	0.213	-0.47	1000	0.2

TABLE 50
Structural Steel > Isotropic Elasticity

Young's Modulus MPa	Poisson's Ratio	Bulk Modulus MPa	Shear Modulus MPa	Temperature C
2.e+005	0.3	1.6667e+005	76923	

TABLE 51
Structural Steel > Isotropic Relative Permeability

Relative Permeability
10000